COPY-PASTE COMMISSIONING PLAYBOOK · OPTION A

Majo Bridge Technologies and Events

# What to do next: LIA physical server + new cloud VM

## Lebawi International Academy · MaJo e-School Bridge (MayaBela)

Version 1.0 · 10 November 2026, East Africa Time (Africa/Nairobi) · For an engineer who has never done this before. Written so you can sit in front of a keyboard and paste.

**Golden rule — read twice.** Never type the shared MaJo address `https://hwkiihonthueadbhcvfi.supabase.co` into either LIA machine, the LIA website build, or the LIA Android APK. Never run `deploy-cloudflare-pages.cmd` for this job. Fnote and other schools stay on `https://mayabela.pages.dev`. LIA gets its own empty database on the school's physical server.

**Option A in one sentence.** The Lebawi physical server holds the only write database (teachers, marks, health). The new cloud VM is the public door (parents on mobile data) plus a disaster-recovery copy. The school router does **not** open ports to the internet. The campus starts an encrypted tunnel **out** to the cloud.

## 0. Picture (so the steps make sense)

```
Parents / phones  --HTTPS-->  CLOUD VM (public IP, padlock)
                                    nginx :443
                                       |
                                       | WireGuard tunnel (campus started it)
                                       v
Staff on school Wi-Fi --LAN-->  CAMPUS SERVER
                                  nginx (LAN) + Kong + Auth + Postgres PRIMARY
                                  nightly backup + USB
                                  replica copy is sent to the cloud for disaster recovery
```

Live logins and saves always go to the campus database. The cloud does not become a second school.

## 1. Who sits where

| Label | Who | Where they type |
|---|---|---|
| OUR SIDE | MaJo engineer | Your laptop + both servers |
| LIA SIDE | LIA IT / principal nominee | Domain login, server room, UPS, admin password |
| CLOUD DESK | Whoever rented the VM (LIA or MaJo on LIA's behalf) | VM dashboard: Ubuntu, public IP, SSH key |

## 2. Write these words on paper first

Do not start until every line has a real value. Then every command below uses those names.

CLOUD_PUBLIC_IP
The new VM's public IPv4

CAMPUS_LAN_IP

School server inside address (example 192.168.1.50)

**YOURDOMAIN.COM**
Domain LIA owns (example lia.edu.et)

**ERP.YOURDOMAIN.COM**
Website teachers and parents open

**API.YOURDOMAIN.COM**
Database API (same two machines)

**SCHOOL_ID**
Short id in the app — letters only (example LIA)

**CLOUDADMIN**
Linux user on the cloud VM

**LIAADMIN**
Linux user on the campus server

**Two hostnames, one public IP — the cloud's IP.** Parents never need the school's public IP. The campus may have no public IP at all. That is OK for Option A.

## 3. Day-by-day map (print this page)

| When | Machine | What you finish |
|---|---|---|
| Day 0 | Office | Paper above filled. Cloud VM exists. Campus has Ubuntu, 16 GB RAM, UPS. |
| Morning 1 | DNS website | `erp` and `api` A records → **CLOUD_PUBLIC_IP**. Grey cloud if Cloudflare. |
| Morning 1 | **CLOUD** | Ubuntu updates, firewall 22/80/443/51820, nginx placeholder. |
| Afternoon 1 | **CAMPUS** | Ubuntu, Docker, LIA's empty Supabase stack, firewall LAN + WireGuard only. |
| Morning 2 | Both | WireGuard. Ping 10.66.0.1 ↔ 10.66.0.2. |
| Morning 2 | **CLOUD** | Nginx proxies API through the tunnel. Certbot padlock. |
| Afternoon 2 | **CAMPUS** | MayaBela SQL + functions. LAN nginx for staff. |
| Morning 3 | Laptop | Flutter web build with LIA URL + LIA anon key. Search JS for the MaJo project ref — must find nothing. |
| Morning 3 | Both | Upload website to campus and cloud. Create only school LIA. Turn off the 10 excluded modules. |
| Afternoon 3 | Both | Isolation tests, backup, copy dump to cloud, USB to principal, paper handover. |

## 4. Phase 0 — Buy / confirm the new cloud VM

**CLOUD DESK** + **OUR SIDE**. Any Ubuntu 22.04 or 24.04 VPS LIA (or MaJo for LIA) controls. Not MaJo's shared Supabase.

| | Ask | Write it |
|---|---|---|
| ☐ | Provider (Ethio / licensed local DC / LIA VPS) | |
| ☐ | Public IPv4 (must not start 10. / 192.168. / 172.16–31.) | |
| ☐ | Size: 2 vCPU, 4 GB RAM minimum (8 GB better), 80 GB disk | |
| ☐ | SSH: you can log in as `cloudadmin` with a key | |
| ☐ | Country/region written down for LIA (agreement Art. 13.2) | |

From your laptop (not school Wi-Fi), test:

```
ping CLOUD_PUBLIC_IP
ssh cloudadmin@CLOUD_PUBLIC_IP
```

You want a login prompt, then a Linux shell. If SSH fails, fix the VM dashboard (firewall, key) before anything else.

## 5. Phase 1 — Point the domain at the *cloud* IP

**BOTH** at the domain registrar. Two A records, **both** values = `CLOUD_PUBLIC_IP` (not the school).

| Type | Name | Value | Proxy |
|---|---|---|---|
| A | `erp` | CLOUD_PUBLIC_IP | DNS only / grey cloud |
| A | `api` | CLOUD_PUBLIC_IP | DNS only / grey cloud |

Wait, then from any network:

```
nslookup erp.yourdomain.com
nslookup api.yourdomain.com
```

Both must show `CLOUD_PUBLIC_IP`. Do not continue to certificates until this is true.

## 6. Phase 2 — First commands on the `CLOUD` VM

SSH in, then paste. Replace nothing yet except if the user is not `cloudadmin`.

```
sudo apt update
sudo apt -y upgrade
sudo apt -y install ca-certificates curl git ufw nginx unzip jq wireguard qrencode
```

```
sudo ufw allow OpenSSH
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw allow 51820/udp
sudo ufw enable
sudo ufw status
```

You want 22, 80, 443, 51820 allowed.

```
sudo mkdir -p /var/www/lia-erp
echo "LIA ERP cloud door — waiting for files" | sudo tee /var/www/lia-erp/index.html
sudo rm -f /etc/nginx/sites-enabled/default
```

## 7. Phase 3 — First commands on the `CAMPUS` physical server

**OUR SIDE** sitting in the school, or AnyDesk. Ubuntu Server 22.04/24.04. 16 GB RAM if possible.

```
ip a
hostname -I
```

Write the LAN address as `CAMPUS_LAN_IP` (usually 192.168.x.x). Tell the school this address must stay fixed in the router DHCP reservation.

```
sudo apt update
sudo apt -y upgrade
sudo apt -y install ca-certificates curl git ufw nginx unzip jq wireguard socat
```

**Campus firewall is stricter.** Do **not** open 80/443 to the whole internet on this box. Staff use the LAN. Parents use the cloud door.

```
sudo ufw allow OpenSSH
sudo ufw allow from 192.168.0.0/16 to any port 80
sudo ufw allow from 192.168.0.0/16 to any port 443
sudo ufw allow 51820/udp
sudo ufw enable
sudo ufw status
```

If the school LAN is 10.x, change `192.168.0.0/16` to `10.0.0.0/8`.

### 7.1 Docker on campus (the database lives here)

```
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]
https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo $VERSION_CODENAME)
stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt update
sudo apt -y install docker-ce docker-ce-cli containerd.io docker-compose-plugin
sudo usermod -aG docker $USER
```

Log out and log in once so Docker works without sudo. Check:

```
docker --version
docker compose version
```

## 8. Phase 4 — WireGuard tunnel (cloud is the hub, campus dials out)

This is the encrypted pipe in the architecture drawing. The campus starts it. The school does not port-forward 443.

### 8.1 On `CLOUD` — make keys

```
sudo mkdir -p /etc/wireguard
cd /etc/wireguard
umask 077
wg genkey | tee server_private.key | wg pubkey > server_public.key
wg genkey | tee campus_private.key | wg pubkey > campus_public.key
sudo cat server_private.key; echo; sudo cat server_public.key; echo
sudo cat campus_private.key; echo; sudo cat campus_public.key
```

Four strings. Label them on paper: CloudPriv, CloudPub, CampusPriv, CampusPub.

```
sudo nano /etc/wireguard/wg0.conf
```

Paste this. Put the real keys and the cloud public IP.

```
[Interface]
Address = 10.66.0.1/24
ListenPort = 51820
PrivateKey = PASTE_CloudPriv
# NAT so campus can be reached as 10.66.0.2
PostUp = iptables -A FORWARD -i wg0 -j ACCEPT; iptables -t nat -A POSTROUTING -o $(ip -4
route get 1.1.1.1 | awk '{print $5; exit}') -j MASQUERADE
PostDown = iptables -D FORWARD -i wg0 -j ACCEPT; iptables -t nat -D POSTROUTING -o $(ip -4
route get 1.1.1.1 | awk '{print $5; exit}') -j MASQUERADE

[Peer]
# campus
PublicKey = PASTE_CampusPub
AllowedIPs = 10.66.0.2/32
```

```
echo "net.ipv4.ip_forward=1" | sudo tee /etc/sysctl.d/99-wireguard.conf
sudo sysctl --system
sudo systemctl enable --now wg-quick@wg0
sudo wg show
```

You want `wg0` listening on 51820. The peer will show “latest handshake” after campus connects.

### 8.2 On `CAMPUS` — dial out

```
sudo nano /etc/wireguard/wg0.conf
```

```
[Interface]
Address = 10.66.0.2/24
PrivateKey = PASTE_CampusPriv

[Peer]
PublicKey = PASTE_CloudPub
Endpoint = CLOUD_PUBLIC_IP:51820
AllowedIPs = 10.66.0.1/32
PersistentKeepalive = 25
```

```
sudo systemctl enable --now wg-quick@wg0
ping -c 3 10.66.0.1
sudo wg show
```

You want replies from 10.66.0.1 and a handshake a few seconds old. If ping fails: cloud UDP 51820, keys swapped wrong, or campus outbound UDP blocked — ask LIA IT to allow UDP 51820 to the cloud IP.

### 8.3 Publish campus Kong only on the tunnel

On **CAMPUS** after Docker/Kong is running (next phase). For now create the helper so it is ready:

```
sudo nano /etc/systemd/system/lia-wg-kong.service
```

```
[Unit]
Description=Expose campus Kong to WireGuard only
After=docker.service wg-quick@wg0.service
Wants=wg-quick@wg0.service

[Service]
ExecStart=/usr/bin/socat TCP-LISTEN:8000,bind=10.66.0.2,fork,reuseaddr TCP:127.0.0.1:8000
Restart=always

[Install]
WantedBy=multi-user.target
```

Do not start this service until Kong answers on campus localhost (Phase 5). Then:

```
sudo systemctl enable --now lia-wg-kong.service
```

## 9. Phase 5 — LIA's empty database on the CAMPUS server

**OUR SIDE**. Same kind of software as MayaBela cloud (Postgres + Auth + Storage + Edge). Brand new empty files. Not a copy of Fnote.

```
sudo mkdir -p /opt/lia
sudo chown $USER:$USER /opt/lia
cd /opt/lia
git clone --depth 1 https://github.com/supabase/supabase.git
cd supabase/docker
cp .env.example .env
```

On your laptop make random strings:

```
openssl rand -base64 32
openssl rand -base64 32
openssl rand -base64 32
```

```
nano /opt/lia/supabase/docker/.env
```

Change at least:

```
SITE_URL=https://erp.yourdomain.com
API_EXTERNAL_URL=https://api.yourdomain.com
SUPABASE_PUBLIC_URL=https://api.yourdomain.com
POSTGRES_PASSWORD=paste-a-long-random-string
```

Leave `ANON_KEY` and `JWT_SECRET` as the matching pair in this LIA `.env`. Copy `ANON_KEY` to a laptop file named `LIA-ONLY-keys.txt`. Never paste the MaJo cloud anon key here.

```
cd /opt/lia/supabase/docker
docker compose pull
docker compose up -d
docker compose ps
```

Wait 1–3 minutes. `kong`, `auth`, `rest`, `storage`, `db` should be running. Then:

```
curl -sS -I http://127.0.0.1:8000/auth/v1/health
sudo systemctl enable --now lia-wg-kong.service
curl -sS -I http://10.66.0.2:8000/auth/v1/health
```

From **`CLOUD`**:

```
curl -sS -I http://10.66.0.2:8000/auth/v1/health
```

You want an HTTP header, not “connection refused”.

## 10. Phase 6 — Public padlock on the `CLOUD` VM

```
sudo nano /etc/nginx/sites-available/lia-erp
```

Paste the whole block. Change the two hostnames only.

```
server {
    listen 80;
    server_name erp.yourdomain.com;
    root /var/www/lia-erp;
    index index.html;
    location / {
        try_files $uri $uri/ /index.html;
    }
}

server {
    listen 80;
    server_name api.yourdomain.com;
    client_max_body_size 50M;
    location / {
        proxy_pass http://10.66.0.2:8000;
        proxy_http_version 1.1;
        proxy_connect_timeout 5s;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
    }
}
```

```
sudo ln -sf /etc/nginx/sites-available/lia-erp /etc/nginx/sites-enabled/lia-erp
sudo nginx -t
sudo systemctl reload nginx
sudo apt -y install certbot python3-certbot-nginx
sudo certbot --nginx -d erp.yourdomain.com -d api.yourdomain.com
```

- Email: principal or ops
- Agree: Y
- Redirect HTTP → HTTPS: choose **2**

Phone on **mobile data** (school Wi-Fi off): open `https://erp.yourdomain.com`. You want “waiting for files” and a closed padlock. If Certbot fails, DNS is still wrong or cloud 80/443 is closed — fix that before building the app.

## 11. Phase 7 — LAN website on the `CAMPUS` server (lessons if the ISP dies)

```
sudo mkdir -p /var/www/lia-erp
sudo chown -R $USER:www-data /var/www/lia-erp
echo "LIA ERP campus — waiting for files" | sudo tee /var/www/lia-erp/index.html
sudo nano /etc/nginx/sites-available/lia-erp
```

```
server {
    listen CAMPUS_LAN_IP:80;
    server_name erp.yourdomain.com CAMPUS_LAN_IP;
    root /var/www/lia-erp;
    index index.html;
    location / {
        try_files $uri $uri/ /index.html;
    }
}

server {
    listen CAMPUS_LAN_IP:80;
    server_name api.yourdomain.com;
    client_max_body_size 50M;
    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
    }
}
```

Replace `CAMPUS_LAN_IP` with the real 192.168 address in both `listen` lines.

```
sudo rm -f /etc/nginx/sites-enabled/default
sudo ln -sf /etc/nginx/sites-available/lia-erp /etc/nginx/sites-enabled/lia-erp
sudo nginx -t
sudo systemctl reload nginx
```

On a staff PC, browser to `http://CAMPUS_LAN_IP` — you should see the campus placeholder. Optional later: copy Let's Encrypt files from the cloud over WireGuard so LAN also has a padlock. Week 1 HTTP-on-LAN is enough if staff normally use the public HTTPS URL.

### 11.1 Optional split-horizon (staff keep working when WAN is down)

On the campus router, if it has "DNS static host": `erp.yourdomain.com` and `api.yourdomain.com` → `CAMPUS_LAN_IP`. Then staff PCs on school Wi-Fi never leave the building. Parents on mobile still hit the cloud IP. If the router cannot do this, add those two lines to each staff PC `hosts` file, or skip and accept that staff also need internet unless they open `http://CAMPUS_LAN_IP`.

## 12. Phase 8 — Load MayaBela onto the campus database

Copy the repo's `supabase` folder onto campus (USB is safest if SSH is not on the internet):

```
ls /opt/lia/mayabela-supabase/migrations
ls /opt/lia/mayabela-supabase/functions
```

You must see `20260725120000_app_documents.sql` and a `school-login` folder.

```
cd /opt/lia/mayabela-supabase
ls migrations | sort
```

Apply oldest first. Repeat the last line for every file, in that order. Stop if one errors.

```
cd /opt/lia/supabase/docker
docker compose exec -T db psql -U postgres -d postgres < /opt/lia/mayabela-
supabase/migrations/20260725120000_app_documents.sql
```

Copy function folders (including `_shared` and `school-login`) into the self-host functions directory this Docker stack uses, then:

```
cd /opt/lia/supabase/docker
docker compose restart functions
```

Studio (school LAN only, never the internet):

```
http://CAMPUS_LAN_IP:54323
```

Create buckets `school-files` (private) and `school-branding` (public). Close 54323 from the world (campus firewall already did).

**Mail and Maya Assistant.** Password-reset mail needs `MAIL_FROM` plus Resend or SMTP secrets on **this** campus stack. Do not set `MAYA_AI_API_KEY` until LIA signs written approval (agreement Art. 3.5).

## 13. Phase 9 — Build the website so it talks only to LIA

**OUR SIDE, WINDOWS LAPTOP.**

**Do not run** `deploy-cloudflare-pages.cmd`.

```
cd C:\path\to\mayabela
flutter pub get
flutter build web --release --pwa-strategy=none --dart-define=SUPABASE_CONFIGURED=true --
dart-define=SUPABASE_URL=https://api.yourdomain.com --dart-
define=SUPABASE_ANON_KEY=PASTE_LIA_ANON_KEY_HERE
```

```
findstr hwkiihonthueadbhcvfi build\web\main.dart.js
```

If that prints any line, the build is wrong. Do not upload it.

Copy `build\web\*` to **both** machines (USB or scp through the tunnel):

- Campus: `/var/www/lia-erp/`
- Cloud: `/var/www/lia-erp/`

```
sudo chown -R www-data:www-data /var/www/lia-erp
```

Phone on mobile data: `https://erp.yourdomain.com` must show the MaJo / MayaBela login, padlock closed. Hard refresh Ctrl+Shift+R.

## 14. Phase 10 — Create only the LIA school

1. Open `https://erp.yourdomain.com` (not pages.dev).
2. Platform console with the **LIA server** owner PIN (this PIN exists only here).
3. Create school ID `LIA`, name Lebawi International Academy.
4. Modules ON = the 49 TOR modules. Modules OFF = `campus`, `cctv`, `finance`, `hr`, `payroll`, `transport`, `transport_buses`, `transport_live_gps`, `add_driver`, `digital_ops`.
5. Create the school-admin account. Log out of platform. Log in as admin. Change the password.

Do not create Fnote or any other school on this campus database.

## 15. Phase 11 — Nightly backup on campus + copy to cloud (the replica)

Week-1 replica for beginners: a daily file copy, not a live read database. All parent traffic still goes through the tunnel to campus. If campus dies, you restore this file onto a spare — you do not fail over onto MaJo shared cloud.

On **CAMPUS**:

```
sudo nano /usr/local/bin/lia-backup.sh
```

```
#!/bin/bash
set -euo pipefail
STAMP=$(date +%F)
DIR=/opt/lia/backups/$STAMP
mkdir -p "$DIR" /opt/lia/backups/latest
cd /opt/lia/supabase/docker
docker compose exec -T db pg_dump -U postgres postgres | gzip > "$DIR/postgres.sql.gz"
cp -f "$DIR/postgres.sql.gz" /opt/lia/backups/latest/postgres.sql.gz
find /opt/lia/backups -mindepth 1 -maxdepth 1 -type d -mtime +14 -exec rm -rf {} \;
```

```
sudo chmod +x /usr/local/bin/lia-backup.sh
sudo crontab -e
```

Add:

```
15 2 * * * /usr/local/bin/lia-backup.sh
40 2 * * * scp /opt/lia/backups/latest/postgres.sql.gz
cloudadmin@10.66.0.1:/opt/lia/replica/postgres.sql.gz
```

On **CLOUD** first:

```
sudo mkdir -p /opt/lia/replica
sudo chown cloudadmin:cloudadmin /opt/lia/replica
```

You need SSH from campus → cloud over WireGuard (key login, no password). Test once by running the backup script by hand. Copy the same file to a USB the principal keeps in another building every week. Restore once to a throwaway folder before you leave — a backup never restored is not a backup.

**Streaming Postgres replica** (hot standby on the cloud) is the “advanced” Option A. Do it after week 1 if LIA wants faster failover. Until then the dump file is the replica. Live reads and writes stay on campus.

## 16. Phase 12 — Prove it (15 minutes, both of you tick)

| | Test | Expected |
|---|---|---|
| ☐ | Parent phone, mobile data: https://erp.yourdomain.com | Login, padlock, LIA branding |
| ☐ | Staff PC on school Wi-Fi: same URL or http://CAMPUS_LAN_IP | Same login |
| ☐ | Log in as LIA admin, add a dummy student | Visible on LIA URL only |
| ☐ | Open https://mayabela.pages.dev as Fnote | Unchanged. Dummy student not there |
| ☐ | Finance / Transport / GPS / CCTV tiles for LIA admin | Hidden or labelled unavailable |
| ☐ | `sudo wg show` on campus | Handshake a few seconds old |
| ☐ | Unplug campus WAN for 30 seconds | Staff on LAN still open CAMPUS_LAN_IP. Parents on mobile wait. pages.dev still lives |
| ☐ | `findstr hwkiihonthueadbhcvfi` on the LIA `main.dart.js` | No lines |

If LIA data appears on the shared cloud, the Flutter build used the wrong URL. Rebuild Phase 9.

## 17. What you hand the school on paper

| | Item | Write it |
|---|---|---|
| ☐ | Public URL (parents) | https://erp.yourdomain.com |
| ☐ | LAN URL (staff if internet is down) | http://CAMPUS_LAN_IP |
| ☐ | School ID | LIA |
| ☐ | “Do not use mayabela.pages.dev for LIA” | On the envelope |
| ☐ | Who reboots campus server / who reboots cloud VM | |
| ☐ | USB backup day | |
| ☐ | MaJo support | +251 911 646 444 |

## 18. If something breaks — first three checks

1. **CAMPUS** `docker compose ps` — all running?
2. **CAMPUS** `sudo wg show` — handshake fresh? If not: `sudo systemctl restart wg-quick@wg0`
3. **CLOUD** `curl -I http://10.66.0.2:8000/auth/v1/health` — if this fails, parents will fail even if the padlock looks fine.

There is **no** automatic fallback onto `www.majobridge.com`. That would copy children's records onto the shared tenant. Restore from last night's dump instead.

## 19. What this is not

- Not copying files onto a Windows desktop and double-clicking
- Not pointing LIA at the current MaJo Supabase project
- Not updating https://mayabela.pages.dev
- Not Option B (cloud as the write database) and not Option C (publish-subset worker)
- Not enabling Finance, Transport, GPS, CCTV, HR, Payroll

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MB-LIA-HYB-A-001 · MaJo e-School Bridge · Option A beginner guide v1.0 · 10 November 2026, Africa/Nairobi.
Companion: Hybrid hosting architecture PDF (Option A). Internal commissioning playbook — not a certificate and not the commercial TOR.